Thus the force on a straight current carrying conductor of length l placed in a uniform magnetic field is

$$\vec{F}_{\text{total}} = \left(I\vec{l} \times \vec{B}\right) \qquad (3.67)$$

In magnitude,

$$F_{\text{total}} = BIl\sin\theta$$

(a) If the conductor is placed along the direction of the magnetic field, the angle then $\theta = 0^{\circ}$. Hence, the force experienced by the conductor is zero.

(b) If the conductor is placed perpendicular to the magnetic field, then the angle $\theta = 90^{\circ}$ Hence, the force experienced by the conductor is maximum, which is $F_{\text{total}} = BIl$.

Fleming's left hand rule

When a current carrying conductor is placed in a magnetic field, the direction of the force experienced by it is given by Fleming's Left Hand Rule (FLHR) as shown in Figure 3.52.

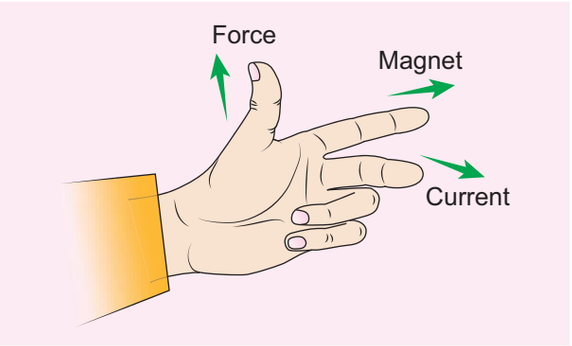
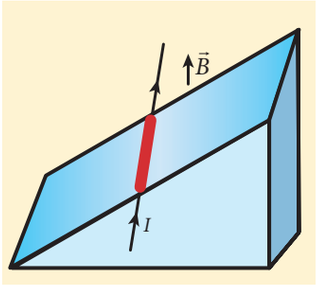


Figure 3.52 Fleming's Left Hand Rule (FLHR)

Stretch out forefinger, the middle finger and the thumb of the left hand such that they are in three mutually perpendicular directions. If the forefinger points in the direction of magnetic field, the middle finger in the direction of the electric current, then thumb will point in the direction of the force experienced by the conductor.

EXAMPLE 3.24

A metallic rod of linear density is 0.25 kg m^{-1} is lying horizontally on a smooth inclined plane which makes an angle of 45° with the horizontal. The rod is not allowed to slide down by flowing a current through it when a magnetic field of strength 0.25 T is acting on it in the vertical direction. Calculate the electric current flowing in the rod to keep it stationary.

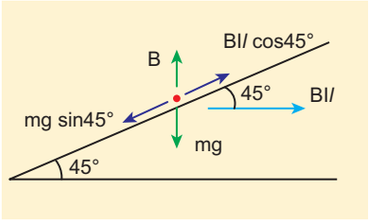


Solution

The linear density of the rod i.e., mass per unit length of the rod is 0.25 kg m^{-1}

$$\Rightarrow \frac{m}{l} = 0.25 \text{ kg m}^{-1}$$

Let I be the current flowing in the metallic rod. The direction of electric current is into the plane of the paper. The direction of magnetic force IBl is given by Fleming's left hand rule.



UNIT-3(XII-Physics_Vol-1).indd 177 14-02-2020 20:04:42